

Aditya Pandey

Software Engineer | Systems · Backend · Infrastructure

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GitHub | LinkedIn | Portfolio

Summary

Software Engineer focused on systems, backend engineering, and distributed architectures. Building Yukti, a distributed data-system project with a custom storage engine, alongside real-time and machine-learning systems. Experienced with Go, Python, JavaScript/TypeScript, Docker, Linux, REST APIs, Socket.IO, and MongoDB.

Technical Skills

Languages: Go, Python, JavaScript, TypeScript, Java

Systems: Distributed Systems, Storage Engines, System Design, Low-Level Design, Networking, Linux

Backend: Node.js, Express.js, REST APIs, Socket.IO

Frontend: React, Next.js, Tailwind CSS

Databases: MongoDB, SQL

Infrastructure & Tools: Docker, Docker Compose, Kubernetes, Git, GitHub, Postman, Bash

Projects

Yukti — Distributed Data System

Systems Engineering · Storage Engine

- Designing a distributed data system around **IDO, UPE, BIL, Discovery, and a custom Storage Engine**, with replication and consensus deferred from the current core.
- Designed Storage Engine architecture around **immutable segments**, including object/segment metadata, read/write paths, versioning, recovery, compaction, and garbage collection.
- Defined low-level storage structures including the **IDO binary format** and rebuildable **Segment Directory**, with explicit metadata ownership and storage-reclamation boundaries.

AIRLINES-INDIGO — Flight Status & Notification System

React · Node.js · MongoDB · Socket.IO · Docker

- Built a containerized full-stack flight-status system using **REST APIs, MongoDB persistence, Socket.IO, Docker, and Docker Compose**.
- Implemented an event pipeline that persists flight-status changes, resolves subscribed passengers, creates notification records, and pushes targeted real-time updates through user-specific Socket.IO rooms.
- Designed persistent notification history and database-level subscription integrity while supporting live dashboard updates and admin-triggered flight events.

Gait Recognition for Border Security

Python · Deep Learning · Computer Vision

- Designed a gait-based person-identification system requiring no conventional biometric input.
- Achieved **87% accuracy** using deep learning.

Cricket Analysis System

Python · Data Processing · Analytics

- Built a ball-by-ball cricket analytics system with bowler-versus-batter performance analysis.
- Developed the data structuring, processing, and analytics pipeline independently.

Experience

ONISORIGINALS

2021 – 2022

Founder / Technical Lead

- Led technical architecture and development of e-commerce and internal software systems.
- Built web applications, backend APIs, automation workflows, and operational dashboards.
- Managed engineering decisions across product development, deployment, and technical operations.

Education

Uttarakhand Technical University, Dehradun

2023 – 2027

B.Tech — Computer Science Engineering

Achievements & Certifications

- **2× National Hackathon Winner**
- **AWS Cloud Practitioner** — ICT Academy
- **NPTEL Certification** — Programming in Java
- **ISRO Hackathon Participant**